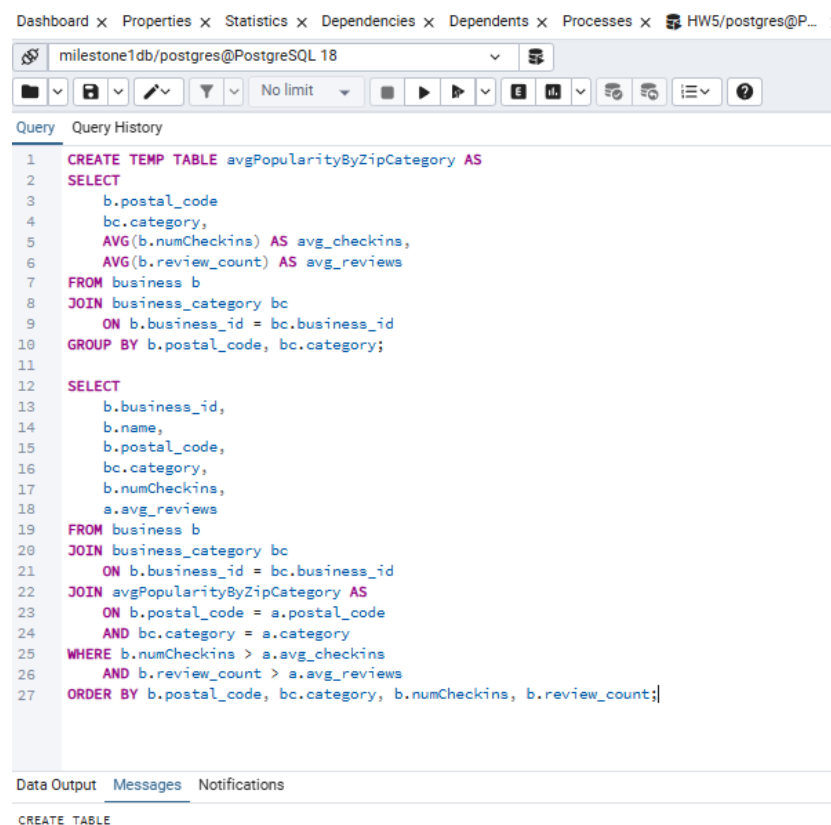


Proposed Business Metrics Paper: Isquared

Popular Businesses Metric:

Description: A business would get classified as popular if it attracts more customers than similar businesses within the same area, that is businesses within the same category and the same zip code. Customer activity would get measured using the total number of check-ins (**numCheckins**) and review count (**review_count**) which would be used to represent how often customers interact with and provide feedback for a business. To ensure fair comparisons, businesses would get evaluated only within the same zipcode and industry, since different regions and categories experience different levels of activity. A business would be considered more popular if the check-ins and review count exceed the average values of businesses in its own zipcode and category.

Query:



```
Dashboard x Properties x Statistics x Dependencies x Dependents x Processes x HWS/postgres@P... :
milestone1db/postgres@PostgreSQL 18
Query History
1 CREATE TEMP TABLE avgPopularityByZipCategory AS
2 SELECT
3     b.postal_code
4     bc.category,
5     AVG(b.numCheckins) AS avg_checkins,
6     AVG(b.review_count) AS avg_reviews
7 FROM business b
8 JOIN business_category bc
9     ON b.business_id = bc.business_id
10 GROUP BY b.postal_code, bc.category;
11
12 SELECT
13     b.business_id,
14     b.name,
15     b.postal_code,
16     bc.category,
17     b.numCheckins,
18     a.avg_reviews
19 FROM business b
20 JOIN business_category bc
21     ON b.business_id = bc.business_id
22 JOIN avgPopularityByZipCategory AS
23     ON b.postal_code = a.postal_code
24     AND bc.category = a.category
25 WHERE b.numCheckins > a.avg_checkins
26     AND b.review_count > a.avg_reviews
27 ORDER BY b.postal_code, bc.category, b.numCheckins, b.review_count;
```

Data Output Messages Notifications

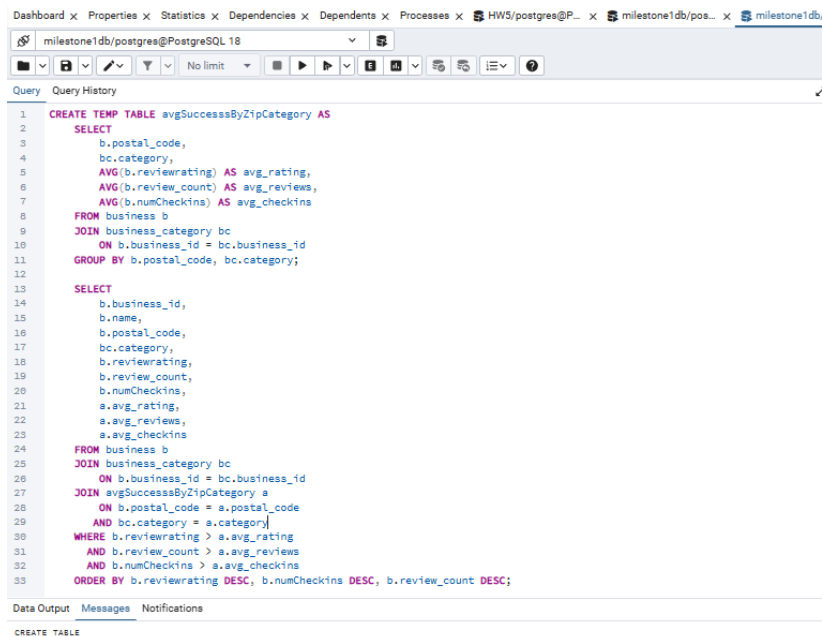
CREATE TABLE

Brief Query Description: This query creates a temporary table that would calculate the average number of check-ins and reviews for each (postal_code, category) pair. Then it would compare each business to these averages and select those whose numCheckins and review_count exceed their groups average. **Tables used:** business, business_category. **Data Items:** postal_code, category, numCheckins, and review_count.

Successful Business Metric:

Description: A business would get classified as successful if it demonstrates excellent customer satisfaction and consistent engagement over time. For a business to classify as “Successful” under our metric, a business would need to have an above average number of the following: **reviewrating**, **review_count**, and **numCheckins**. Again, to ensure fairness when comparing businesses, these would have to get evaluated within the same **postal_code** and **category**.

Query:



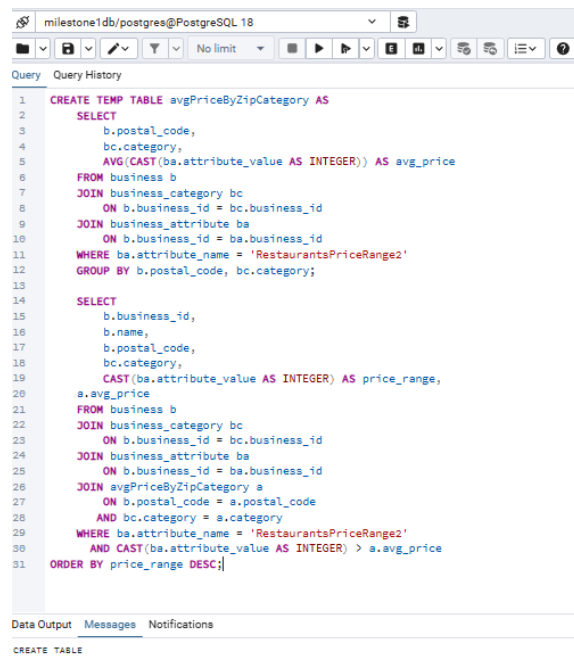
```
1 CREATE TEMP TABLE avgSuccessByZipCategory AS
2 SELECT
3     b.postal_code,
4     bc.category,
5     AVG(b.reviewrating) AS avg_rating,
6     AVG(b.review_count) AS avg_reviews,
7     AVG(b.numCheckins) AS avg_checkins
8 FROM business b
9 JOIN business_category bc
10 ON b.business_id = bc.business_id
11 GROUP BY b.postal_code, bc.category;
12
13 SELECT
14     b.business_id,
15     b.name,
16     b.postal_code,
17     bc.category,
18     b.reviewrating,
19     b.review_count,
20     b.numCheckins,
21     a.avg_rating,
22     a.avg_reviews,
23     a.avg_checkins
24 FROM business b
25 JOIN business_category bc
26 ON b.business_id = bc.business_id
27 JOIN avgSuccessByZipCategory a
28 ON b.postal_code = a.postal_code
29 AND bc.category = a.category
30 WHERE b.reviewrating > a.avg_rating
31 AND b.review_count > a.avg_reviews
32 AND b.numCheckins > a.avg_checkins
33 ORDER BY b.reviewrating DESC, b.numCheckins DESC, b.review_count DESC;
```

Brief Query Description: This query would create a temporary table that calculates the average rating, review count, and check-ins for each (postal_code, category) pair. Then it would compare each business to these averages and select those whose **reviewrating**, **review_count**, and **numCheckins** are above their groups average. **Tables used:** business, business_category. **Data Items:** postal_code, category, reviewrating, review_count, numCheckins.

Expensive Business Metric:

Description: A business would get classified as expensive if its price range exceeds the average price range of other businesses within the same category and zipcode. Expense is measured using the **RestaurantsPriceRange1** attribute from the **business_attribute** table, as well as with **postal_code** and category information to make sure comparisons are fair. A business would be considered as more expensive if its price range value is above the average of its peers within the above parameters.

Query:



```
1 CREATE TEMP TABLE avgPriceByZipCategory AS
2 SELECT
3     b.postal_code,
4     bc.category,
5     AVG(CAST(ba.attribute_value AS INTEGER)) AS avg_price
6 FROM business b
7 JOIN business_category bc
8     ON b.business_id = bc.business_id
9 JOIN business_attribute ba
10     ON b.business_id = ba.business_id
11 WHERE ba.attribute_name = 'RestaurantsPriceRange2'
12 GROUP BY b.postal_code, bc.category;
13
14 SELECT
15     b.business_id,
16     b.name,
17     b.postal_code,
18     bc.category,
19     CAST(ba.attribute_value AS INTEGER) AS price_range,
20     a.avg_price
21 FROM business b
22 JOIN business_category bc
23     ON b.business_id = bc.business_id
24 JOIN business_attribute ba
25     ON b.business_id = ba.business_id
26 JOIN avgPriceByZipCategory a
27     ON b.postal_code = a.postal_code
28     AND bc.category = a.category
29 WHERE ba.attribute_name = 'RestaurantsPriceRange2'
30     AND CAST(ba.attribute_value AS INTEGER) > a.avg_price
31 ORDER BY price_range DESC;
```

Data Output Messages Notifications

CREATE TABLE

Brief Query Description: This query creates a temporary table that calculates the average price range for each (postal_code, category) pair using the RestaurantsPriceRange2 attribute. It would then compare each business to these averages and select those whose price is above the average of their peers within the above parameters. **Tables Used:** business, business_category, business_attribute. **Data Items:** postal_code, category, and RestaurantsPriceRange2.